MikroTik MPLS L3VPN Lab

After several days of design, configuration, troubleshooting, and testing, I am pleased to share the successful completion of my MikroTik MPLS L3VPN Lab built entirely in GNS3.

Technologies Implemented

✅ IS-IS as the Core IGP
✅ MPLS & LDP Label Distribution
✅ MP-BGP VPNv4
✅ Route Reflector Architecture
✅ VRF (Virtual Routing and Forwarding)
✅ eBGP PE-CE Connectivity
✅ AS Override for Customer Sites using the same AS
✅ Centralized Internet Breakout (PE1)
✅ Default Route Propagation Across the MPLS VPN
✅ NAT and Internet Access for all Branches

Network Overview

The lab consists of:

- 4 Provider Edge (PE) Routers
- 4 Core Provider (P) Routers
- 1 Route Reflector (RR)
- 6 Branch Offices
- 1 Headquarters Site
- Customer AS: 65520
- Provider AS: 65500

All customer sites are interconnected through an MPLS backbone while maintaining routing separation using a dedicated **BANK** VRF.

Key Learning Outcomes

One of the most interesting challenges during the implementation was troubleshooting VPN route propagation between PE and CE routers.

A critical discovery was that in MikroTik RouterOS v7, VPN-imported routes are classified as: *bgp-mpls-vpn* and must be explicitly redistributed on PE-to-CE BGP sessions using:
Output Redistribute:
✓ bgp
✓ bgp-mpls-vpn

Without this configuration, VPN routes are successfully imported into the VRF but are not advertised to customer edge routers.

Another valuable lesson was implementing centralized Internet access through a designated Internet Gateway PE and propagating a default route throughout the MPLS VPN.

Final Results

- Full branch-to-branch connectivity across all sites
- End-to-end MPLS VPN routing operational
- Centralized Internet access working for all branches
- Successful communication between all customer LANs
- VPN route reflection functioning correctly via Route Reflector

This project provided hands-on experience with technologies commonly deployed by Service Providers and large Enterprise WAN environments.

Continuous learning and practical labs remain one of the best ways to deepen networking knowledge and troubleshooting skills.

Lab Status: ✅ Fully Operational 🎉

IP ADDRESSING

P1				P3			
ADDRESS	**NETWORK**	**INTERFACE**	**VRF**	**ADDRESS**	**NETWORK**	**INTERFACE**	**VRF**
10.255.0.11/32	10.255.0.11	Loopback	Main	10.255.0.13/32	10.255.0.13	Loopback	Main
10.0.1.1/30	10.0.1.0	Ether2	Main	10.0.4.2/30	10.0.4.0	Ether1	Main
10.0.4.1/30	10.0.4.0	Ether3	Main	10.0.3.1/30	10.0.3.0	Ether2	Main
10.0.5.1/30	10.0.5.0	Ether4	Main	10.0.6.2/30	10.0.6.0	Ether3	Main
10.0.7.1/30	10.0.7.0	Ether5	Main	10.0.9.1/30	10.0.9.0	Ether4	Main
10.0.10.1/30	10.0.10.0	Ether6	Main	10.0.8.1/30	10.0.8.0	Ether5	Main
				10.0.21.1/30	10.0.21.0	Ether6	Main

P2				P4			
ADDRESS	**NETWORK**	**INTERFACE**	**VRF**	**ADDRESS**	**NETWORK**	**INTERFACE**	**VRF**
10.255.0.12/32	10.255.0.12	Loopback	Main	10.255.0.14/32	10.255.0.14	Loopback	Main
10.0.1.2/30	10.0.1.0	Ether2	Main	10.0.2.2/30	10.0.2.0	Ether1	Main
10.0.2.1/30	10.0.2.0	Ether3	Main	10.0.3.2/30	10.0.3.0	Ether2	Main
10.0.6.1/30	10.0.6.0	Ether4	Main	10.0.5.2/30	10.0.5.0	Ether3	Main
10.0.23.1/30	10.0.23.0	Ether5	Main	10.0.24.1/30	10.0.24.0	Ether4	Main
0.0.14.1/30	10.0.14.0	Ether6	Main	10.0.13.1/30	10.0.13.0	Ether5	Main
				10.0.22.1/30	10.0.22.0	Ether6	Main

PE1				PE3			
ADDRESS	**NETWORK**	**INTERFACE**	**VRF**	**ADDRESS**	**NETWORK**	**INTERFACE**	**VRF**
10.255.0.1/32	10.255.0.1	Loopback	Main	10.255.0.3/32	10.255.0.3	Loopback	Main
10.0.7.2/30	10.0.7.0	Ether1	Main	10.0.9.2/30	10.0.9.0	Ether1	BANK
10.0.5.1/30	10.0.5.0	Ether2	BANK	10.0.11.1/30	10.0.11.0	Ether2	BANK
10.0.12.1/30	10.0.12.0	Ether3	BANK	10.0.19.1/30	10.0.19.0	Ether3	Main
10.0.8.2/30	10.0.8.0	Ether4	Main	10.0.10.2/30	10.0.10.0	Ether4	Main
10.0.20.2/30	10.0.20.0	Ether5	BANK				
192.168.118.134/24	**192.168.118.0**	**Ether6**	BANK				

PE2

ADDRESS	NETWORK	INTERFACE	VRF
10.255.0.2/32	10.255.0.2	Loopback	Main
10.0.23.2/30	10.0.23.0	Ether1	Main
10.0.17.1/30	10.0.17.0	Ether2	BANK
10.0.13.2/30	10.0.13.0	Ether3	Main
10.0.13.2/30	10.0.13.0	Ether4	BANK
10.0.16.1/30	10.0.16.0	Ether5	BANK

PE4

ADDRESS	NETWORK	INTERFACE	VRF
10.255.0.4/32	10.255.0.4	Loopback	Main
10.0.24.2/30	10.0.24.0	Ether1	Main
10.0.15.1/30	10.0.15.0	Ether2	BANK
10.0.18.1/30	10.0.18.0	Ether3	BANK
10.0.14.2/30	10.0.14.0	Ether4	Main

CE1

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.1/32	10.254.0.1	Loopback	Main
10.0.12.2/30	10.0.12.0	Ether1	Main
192.168.201.1/24	192.168.201.0	Ether2	Main

CE2

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.2/32	10.254.0.2	Loopback	Main
10.0.5.2/30	10.0.5.0	Ether1	Main
10.0.19.2/30	10.0.19.0	Ether2	Main
192.168.202.1/24	192.168.202.0	Ether3	Main

CE3

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.3/32	10.254.0.3	Loopback	Main
10.0.11.2/30	10.0.11.0	Ether1	Main
192.168.203.1/24	192.168.203.0	Ether2	Main

CE4

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.4/32	10.254.0.4	Loopback	Main
10.0.17.2/30	10.0.17.0	Ether1	Main
192.168.204.1/24	192.168.204.0	Ether2	Main

CE5

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.5/32	10.254.0.5	Loopback	Main
10.0.16.2/30	10.0.16.0	Ether1	Main
10.0.15.2/30	10.0.15.0	Ether2	Main
192.168.205.1/24	192.168.205.0	Ether3	Main

CE6

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.6/32	10.254.0.6	Loopback	Main
10.0.18.2/30	10.0.18.0	Ether1	Main
192.168.206.1/24	192.168.206.0	Ether2	Main

HQ

ADDRESS	NETWORK	INTERFACE	VRF
10.254.0.7/32	10.254.0.7	Loopback	Main
10.0.20.1/30	10.0.20.0	Ether2	Main
10.0.25.1/30	10.0.25.0	Ether3	Main
192.168.207.1/24	192.168.207.0	Ether4	Main

PC3

ADDRESS	NETWORK
192.168.203.10/24	192.168.203.0

PC4

ADDRESS	NETWORK
192.168.204.10/24	192.168.204.0

PC1

ADDRESS	NETWORK
192.168.201.10/24	192.168.201.0

PC5

ADDRESS	NETWORK
192.168.205.10/24	192.168.205.0

PC2

ADDRESS	NETWORK
192.168.202.10/24	192.168.202.0

PC6

ADDRESS	NETWORK
192.168.206.10/24	192.168.206.0

Branch to Branch Reachability Test

Can PC1 ping PC 2? YES

PC1> ping 192.168.202.10 [CE 2 LAN)

```
PC1> ping 192.168.202.10

84 bytes from 192.168.202.10 icmp_seq=1 ttl=61 time=20.854 ms
84 bytes from 192.168.202.10 icmp_seq=2 ttl=61 time=6.857 ms
84 bytes from 192.168.202.10 icmp_seq=3 ttl=61 time=9.877 ms
84 bytes from 192.168.202.10 icmp_seq=4 ttl=61 time=3.890 ms
84 bytes from 192.168.202.10 icmp_seq=5 ttl=61 time=3.241 ms
```

Can PC1 ping PC 3? YES

PC1> ping 192.168.202.10 [CE 3 LAN)

```
PC1> ping 192.168.203.10

84 bytes from 192.168.203.10 icmp_seq=1 ttl=60 time=11.321 ms
84 bytes from 192.168.203.10 icmp_seq=2 ttl=60 time=7.057 ms
84 bytes from 192.168.203.10 icmp_seq=3 ttl=60 time=6.270 ms
84 bytes from 192.168.203.10 icmp_seq=4 ttl=60 time=4.525 ms
84 bytes from 192.168.203.10 icmp_seq=5 ttl=60 time=5.790 ms
```

Can PC6 ping PC 1? YES [CE 1 LAN)

```
PC6> ping 192.168.201.10

84 bytes from 192.168.201.10 icmp_seq=1 ttl=60 time=8.571 ms
84 bytes from 192.168.201.10 icmp_seq=2 ttl=60 time=6.521 ms
84 bytes from 192.168.201.10 icmp_seq=3 ttl=60 time=7.699 ms
84 bytes from 192.168.201.10 icmp_seq=4 ttl=60 time=5.728 ms
84 bytes from 192.168.201.10 icmp_seq=5 ttl=60 time=4.571 ms
```

Can PC5 ping PC 2? YES [CE 2 LAN)

```
PC5> ping 192.168.202.10

84 bytes from 192.168.202.10 icmp_seq=1 ttl=60 time=8.796 ms
84 bytes from 192.168.202.10 icmp_seq=2 ttl=60 time=3.684 ms
84 bytes from 192.168.202.10 icmp_seq=3 ttl=60 time=4.064 ms
84 bytes from 192.168.202.10 icmp_seq=4 ttl=60 time=8.030 ms
84 bytes from 192.168.202.10 icmp_seq=5 ttl=60 time=4.099 ms
```

INTERNET ACCESS TEST

From CE1, PC1

```
PC1> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=126 time=14.566 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=126 time=16.156 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=126 time=16.409 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=126 time=13.789 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=126 time=15.041 ms
```

From CE6, PC6

```
PC6> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=125 time=18.451 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=125 time=19.860 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=125 time=16.394 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=125 time=19.045 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=125 time=18.467 ms
```

From HQ, PC7

```
PC7> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=126 time=15.321 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=126 time=14.463 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=126 time=15.624 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=126 time=18.428 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=126 time=18.390 ms
```

For Consultation

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